

Abdelrahman Wael Mohamed

Artificial Intelligence Engineer

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Summary

Artificial Intelligence Engineer with expertise in machine learning, deep learning, computer vision, generative AI, large language models (LLMs), retrieval-augmented generation (RAG), and agentic AI. Passionate about developing intelligent systems that leverage modern AI techniques to solve real-world problems.

Education

B.Sc. in Artificial Intelligence

Faculty of Informatics and Computer Science

British University in Egypt (BUE)

2022 – 2026

Dual Degree — First Class (UK) | Very Good with Honours (Egyptian)

Experience

Machine Learning Engineering Intern | DEPI Program

Oct 2024 - June 2025

- Completed an intensive Machine Learning Engineering internship with hands-on training in statistics, linear algebra, Python, data preprocessing, machine learning, deep learning, NLP, and computer vision.
- Trained and evaluated machine learning models for regression, classification, and clustering, focusing on performance optimization and accuracy improvement.

Generative AI Professional | DEPI Program

Nov 2025 – Jul 2026

- Completed intensive training in Generative AI, Large Language Models (LLMs), and Agentic AI systems.
- Developed Retrieval-Augmented Generation (RAG) systems for knowledge retrieval and question answering.
- Explored multi-agent architectures, AI deployment, monitoring, and responsible AI practices.
- Worked with LLM APIs, prompt engineering techniques, and open-source AI models.

Projects

Hybrid-VTON: Diffusion-Based Virtual Try-On System (Graduation Project) ([View Project](#))

- Developed a Stable Diffusion XL-based virtual try-on framework supporting realistic upper- and lower-body garment transfer while preserving human identity.
- Designed a hybrid alignment framework combining explicit geometric matching with implicit garment-person feature interaction through ClothNet, IP-Adapter, EMASC, and dual GMMs.
- Trained and evaluated on a unified VITON-HD + DressCode-MR dataset and achieving competitive performance after only 35 training epochs, compared with 100+ epochs reported for other frameworks.
- Outperformed several state-of-the-art VTON frameworks on LPIPS (0.077) and FID (6.27).

AI Fashion Designer: Multi-Agent Fashion Image Generation System ([View Project](#))

- Developed an end-to-end fashion generation system integrating a conversational multi-agent framework with a fine-tuned Stable Diffusion XL model.
- Built an automated Qwen2.5-VL captioning and preprocessing pipeline for fashion image-text data used to fine-tune SDXL.
- Designed a LangGraph-based multi-agent workflow for clothing attribute extraction, conversational memory, prompt generation, and image generation.

Healthcare RAG System ([View Project](#))

- Developed a hypertension-focused RAG chatbot for evidence-based medical question answering with source attribution.
- Implemented hybrid retrieval using BGE-M3, BM25, weighted RRF, and BGE reranking, followed by Qwen3-4B for grounded answer generation.

Certificates

- AI & Data Science -Microsoft Machine Learning Engineer from DEPI.
 - Introduction to Machine Learning from AITB.
 - Supervised Machine Learning: Regression and Classification from DeepLearning.AI.
 - Advanced Learning Algorithms from DeepLearning.AI.
 - Python Programming fundamentals from Mahara Tech.
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Skills

- **Programming Languages:** Python, C++, Java, SQL.
 - **Machine Learning & AI:** Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, LLMs, RAG, Agentic AI.
 - **Libraries & Frameworks:** PyTorch, TensorFlow, Scikit-Learn, Hugging Face, OpenCV, LangChain, LangGraph.
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Languages

- English (B2). - Arabic (Native)